Generalised Langford sequences as a CSP in MiniZinc

Symbolic Explainable AI 2026/27 Group 20: António Delgado (106658), Antero Morgado (119213)

CSPLib problem 024 (Langford's number problem)

Video: `https://drive.google.com/drive/folders/1JhtKAuOsx3BNYTG6GkInrLacWVWrHRSs?usp=sharing`

1 Problem

Two copies of each number $1..n$ must be placed in a sequence so that the two copies of i are $i+1$ positions apart. Our starting point is the CSPLib model by Hakan Kjellerstrand (`langford2.mzn`), which only supports this classic case.

2 Extension

Our model (`langford_gen.mzn`) generalises the problem to s copies of each number, with consecutive copies of i exactly $i+d$ positions apart ($s=2, d=1$ is the classic problem, $d=0$ the Nickerson variant). The decision variables are only the positions of the first copies, so the spacing always holds and the search works on n variables. We added reversal symmetry breaking, which halves the search, and a parity constraint for the classic case, which rejects impossible instances ($n \not\equiv 0, 3 \pmod 4$) before search: $n = 13$ takes 0.06 s instead of 52 s with the baseline. When there is no solution, `proj.py` also explains why in plain words. All results were checked against a brute-force enumerator; the benchmark is in `results.csv`.

3 Running the code

Requirements: Python 3.11.2, MiniZinc 2.10.1 (Gecode 6.4.0, Chuffed 0.14.0) on the `PATH`, and the `minizinc` Python package 0.10.0. Keep both `.mzn` files next to `proj.py`.

```
python3 proj.py 8                    # first solution for n=8
python3 proj.py 9 -s 3 --all         # all solutions with 3 copies (d=1)
python3 proj.py 14 --parity          # impossible instance, with explanation
python3 proj.py --selftest           # compare with brute force
python3 proj.py --benchmark          # writes results.csv
```

4 Use of generative AI and web resources

We used generative AI tools only to generate part of the code and to debug. The modelling and solving of the problem were done by us. For background we consulted dialectrix.com/langford.html, dialectrix.com/langford/langford-biblio.html, MathWorld and lclark.edu/∼miller/langford (archived).